

Saif Shikalgar

Automotive Cybersecurity & Embedded Security Engineer

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Education

G. H. Raisoni College of Engineering and Management

Pune, India

B.Tech in Computer Science & Engineering (Cybersecurity Specialization); **CGPA: 8.10/10.0**

Sep 2023 – May 2027 (Expected)

Coursework: Automotive Cybersecurity, Network Security & Cryptography, Operating Systems, Microcontrollers & Embedded Systems, Data Structures, Threat Modeling & Vulnerability Assessment.

Technical Skills

Automotive Protocols & Security: CAN Bus 2.0B / CAN-FD, ISO 11898, OBD-II (J1979), TWAI, ISO/SAE 21434 Concepts, UN R155/R156, Threat Analysis & Risk Assessment (TARA), SecOC, Hardware Firewalls, CAN IDS/IPS.

Embedded Systems & Hardware: STM32 (HAL/LL/CMSIS), ARM Cortex-M4, ESP32, FreeRTOS, Bare-Metal C/C++, Microsecond Timing, Hardware Interrupts (ISR), MCP2515, Logic Analyzers, Oscilloscopes, SPI, I2C, UART.

Security Tools & Analysis: Wireshark, CANalyzer/CANoe concepts, SavvyCAN, Metasploit, Exploit-DB, Packet Fuzzing, Jitter Statistical Analysis, Mahalanobis Distance Engine, Reverse Engineering.

Programming & Platforms: C, C++ (C11/C++20), Python, Bash, Linux Kernel Networking, Docker, NVIDIA Jetson Nano/Orin.

Work Experience & Engineering Leadership

Team Abhyuday Racing

Pune, India

Technical Lead & Department Head – Drive-by-Wire (Automotive R&D)

Jan 2025 – Present

- Designed and deployed a distributed **3-ECU network (NVIDIA Jetson Nano + STM32F446RE + ESP32)** over CAN 2.0B bus with hardware heartbeats, priority message arbitration, and 99.9% packet reliability under severe EMI noise.
- Programmed real-time deterministic **Autonomous Emergency Braking (AEB)** control logic, securing **1st Place Nationally at aBAJA 2026** by halting safely at **6.2m** at 30 km/h with **<12ms CAN reaction latency**.
- Spearheaded an emergency **8-hour overnight hardware migration** from damaged STM32 to ESP32 FreeRTOS architecture at the racetrack, winning **1st Place in Adaptive Cruise Control (ACC)** at aBAJA 2025.
- Engineered a modular Brake-by-Wire pneumatic actuator system with AS5600 12-bit magnetic angle encoders and pressure transducers, cutting hardware costs by **50%** and winning the **1st Place Manufacturing Excellence Award**.

AuraByte Studios (Independent Venture)

Pune, India

Lead Systems Developer & Founder

Dec 2025 – Present

- Architected secure data handling mechanisms with atomic persistence and cryptographic token verification for over **5.73K+ total installs and 2.09K+ active devices**.
- Implemented secure cloud communication pipelines using HTTPS, REST APIs, and GCP serverless infrastructure for license verification and webhook processing.

Flagship Automotive Security Projects

Dual-Gate STM32 Hardware CAN Firewall & Anomaly IDS/IPS | STM32F446RE, C11, CMSIS/HAL, FreeRTOS, CAN 2.0B

- Engineered an inline dual-transceiver CAN bus firewall intercepting OBD-II injection, fuzzing, and RPM spoofing attacks with a deterministic **5.92 μ s processing latency** (1,065 clock cycles at 180 MHz).
- Implemented fixed-point Mahalanobis distance anomaly detection and jitter statistical analysis, rejecting **88.5% of noise** and achieving a **99%+ threat mitigation rate** with zero bus contention.
- Integrated an isolated secondary CAN transceiver pipeline with automatic hardware fail-safe bypass in the event of microcontroller power disruption.

Automotive Telematics & 24S BMS Security Monitor | ESP32, FreeRTOS, BLE GATT, Python, C++

- Reverse-engineered industrial JK BMS BLE GATT protocols to inspect encrypted telemetry payloads across a 76.8V 24S LiFePO4 battery pack, identifying unauthenticated GATT read vulnerabilities.
- Implemented real-time anomaly alerts for unauthorized parameter modification and abnormal cell voltage deltas (<15mV) with hardware contactor cutoff.

Suraksha360: Hardware Security Scanner | C++, ESP8266, Arduino, Exploit-DB, Windows Registry API

- Created a plug-and-play USB hardware penetration testing tool that auto-executes non-destructive vulnerability audits on host machines, cross-referencing software configurations against Exploit-DB for known CVEs.

Honors & Awards

1st Place, Autonomous Emergency Braking (AEB): National aBAJA 2026 Competition; achieved a 6.2m halt on a 6.0m autonomous target threshold at 30 km/h.

1st Place, Manufacturing Excellence Award: National aBAJA 2026 Competition; recognized for modular Drive-by-Wire and Brake-by-Wire actuator design.

1st Place, Adaptive Cruise Control (ACC): aBAJA 2025; achieved following an 8-hour high-stakes crisis hardware migration from STM32 to ESP32.

1st Runner Up, MATLAB Advanced Simulation: National aBAJA 2026 Competition; developed a GPS-based autonomous traversal algorithm.